Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/26/2012
PI Name:	Stutte, Gary Ph.D.		
Project Title:	Photosynthesis and Metabolism of Super Dwarf Wheat in Microgravity (PESTO)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	QinetiQ-North America/NASA Kennedy Space Center		
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PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:	NOTE: PI had a change of employment from Dynamac Corporation to QinetiQ-North America, effective 1 March, 2011.		
Project Type:	Flight	**Solicitation / Funding Source:**	96-OLMSA-01
Start Date:	10/22/1997	**End Date:**	09/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-110; ISS 8A		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC10-0027		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The photosynthetic rate of higher plants is a critical component of plant-based atmospheric regeneration systems being proposed for long-duration space missions. It is only through direct measurement of photosynthesis in microgravity that an informed decision on the feasibility and design of these systems can be made. The overall objective of this research is to determine the effect of microgravity on photosynthetic response of plant tissues developed in either gravity or microgravity. The specific objectives of this research are: 1) Determine the effect of microgravity on CO2 and light response curves of wheat. 2) Determine the effect of microgravity on metabolism and electron transport processes associated with photosynthetic and respiratory gas exchange. 3) Determine the effect of microgravity on carbohydrate partitioning in wheat. 4)Determine the effect of microgravity on gas exchange, including H2O, over range of atmospheric vapor pressure deficits. 5) Utilize the knowledge gained to understand the response of plants grown under		

Task Description:	elevated CO2 conditions of commercial controlled environment crop production systems. To test the hypothesis that net carbon exchange rates are sustained in microgravity, we propose to measure and characterize the CO2 and light response curves for a wheat photosynthetic canopy grown in microgravity. Net carbon exchange and water movement through the canopy will also be measured over defined range of vapor pressure deficits. Post-flight analysis of the tissue for primary photosynthesis parameters, including PSI, PSII, electron transport, and carbohydrate partitioning, will be made and correlated to in-flight data. These measurements of whole canopy gas exchange in microgravity will be used to understand the effects of microgravity on photosynthesis, to quantify the effects on metabolism, and to model the impact of microgravity on biological approaches to atmospheric regeneration for long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will increase our understanding of plant growth and development as affected by space flight conditions. The monitoring of gas exchange will provide direct measurements of the fundamental processes of photosynthesis and transpiration critical to life support on long-duration space missions. Taken together, the three dimensions of the experiment (gas exchange, partitioning, and metabolism) will result in a more complete picture of microgravity influences on photosynthesis. The CO2 and light response curves will allow researchers to establish whether canopy photosynthetic responses are affected by space conditions. This is significant since plants can be used to regenerate the atmosphere in space conditions though removal of potentially toxic CO2 and production of O2. In addition, the tests to evaluate movement of H2O via transpiration are important since they are indicative of stomatal responses which regulate photosynthesis. Further, the impact of microgravity on transpiration is significant since plants can be used to purify water under space flight conditions. These studies will demonstrate the suitability of various solid substrates for supporting plant growth which can be transferred to commercial activities. These studies involving gas exchange at elevated CO2 concentrations increase our understanding of biological impacts of increasing levels of atmospheric CO2 on Earth-based ecosystems. Further, our levels of understanding of plant responses under range of CO2 and light conditions have potential benefits to commercial controlled environment agriculture industries.
Task Progress:	[Ed. note 3/26/2012: compiled from publication list sent by PI and abstract from relevant publication; no specific Task Book report submitted by PI] Plant stand gas exchange was measured nondestructively in microgravity during the Photosynthesis Experiment Subsystem Testing and Operations experiment conducted onboard the International Space Station. Rates of evapotranspiration and photosynthesis measured in space were compared with ground controls to determine if microgravity directly affects whole-stand gas exchange of Triticum aestivum. During six 21-day experiment cycles, evapotranspiration was determined continuously from water addition rates to the nutrient delivery system, and photosynthesis was determined from the amount of CO2 added to maintain the chamber CO2 concentration setpoint. Plant stand evapotranspiration, net photosynthesis, and water use efficiency were not altered by microgravity. Although leaf area was significantly reduced in microgravity-grown plants compared to ground control plants, leaf area distribution was not affected enough to cause significant differences in the amounts of light absorbed by the flight and ground control plant stands. Microgravity also did not affect the response of evapotranspiration to changes in chamber vapor pressure difference of 12-day-old wheat plant stands. These results suggest that gravity naïve plants grown at moderate light levels (300 mgr mol m–2 s–1) behave the same as ground control plants. This implies that future plant-based regenerative life support systems can be sized using 1 g data because water purification and food production rates operate at nearly the same rates as in 1 g at moderate light levels. However, it remains to be verified whether the present results are reproducible in plants grown under stronger light levels.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Articles in Peer-reviewed Journals	Monje O, Anderson S, Stutte GW. "The effects of elevated root zone temperature on the development and carbon partitioning of spring wheat." Journal of the American Society for Horticultural Science 2007 Mar;132(2):178-84. http://journal.ashspublications.org/content/132/2/178.full , Mar-2007
Articles in Peer-reviewed Journals	Stutte GW, Monje O, Hatfield RD, Paul A-L, Ferl RJ, Simone CG. "Microgravity effects on leaf morphology, cell structure, carbon metabolism and mRNA expression of dwarf wheat." Planta. 2006 Oct;224(5):1038-49. Epub 2006 May 10. PubMed PMID: 16708225 , Oct-2006
Articles in Peer-reviewed Journals	Paul A-L, Levine HG, McLamb W, Norwood KL, Reed D, Stutte GW, Wells HW, Ferl RJ. "Plant molecular biology in the space station era: utilization of KSC fixation tubes with RNAlater." Acta Astronaut. 2005 Mar;56(6):623-8. PubMed PMID: 15736319 ; http://dx.doi.org/10.1016/j.actaastro.2004.10.001 , Mar-2005
Articles in Peer-reviewed Journals	Monje O, Stutte G, Chapman D. "Microgravity does not alter plant stand gas exchange of wheat at moderate light levels and saturating CO2 concentration." Planta. 2005 Oct;222(2):336-45. Epub 2005 Jun 21. PubMed PMID: 15968511 , Oct-2005
Articles in Peer-reviewed Journals	Stutte GW, Monje O, Goins GD, Tripathy BC. "Microgravity effects on thylakoid, single leaf, and whole canopy photosynthesis of dwarf wheat." Planta. 2005 Dec;223(1):46-56. Epub 2005 Sep 14. PubMed PMID: 16160842 , Dec-2005
Articles in Peer-reviewed Journals	Liao J, Liu G, Monje O, Stutte GW, Porterfield DM. "Induction of hypoxic root metabolism results from physical limitations in O2 bioavailability in microgravity." Adv Space Res. 2004;34(7):1579-84. PMID: 15880895 ; http://dx.doi.org/10.1016/j.asr.2004.02.002 , Mar-2004